



Classification and Dimension Reduction on MNIST

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BIG Data
16/4/2026



Goal

This project studies classification on a subset of the MNIST handwritten digit dataset.

Different methods with different values for variable were compared to determine which one was best at classifying MNIST data.



Method

1. MNIST dataset was installed
2. Dataset was split into training, testing and validation
3. Data was scaled to ensure training captures the structure of the data and not difference in scale for different features
4. SVD was used on the training data, splitting it into three matrices
5. This was then used to reduce the data to only the most important SVD components
 - a. To reduce dimensionality which allows us to remove noise and improve generalisation
6. A classifier was trained with certain values for its variables
7. Step 5-6 was repeated for different classifiers with different values
8. Step 3-7 was repeated 10 times
9. The classifier classifiers and variable values with the lowest test validation were identified and their test error was compared.



Models compared:

Kneighbours classifier

K values: [1,2,5,15]

Number of neighbours we train on

P values: [2, 10, 25, 50, 100]

Number of SVD components we train on

Random Forest classifier

N values: [10, 50, 100, 200]

Number of trees we train on

P values: [2, 10, 25, 50, 100]

Number of SVDcomponents we train on

Logistic classifier

P values: [2, 10, 25, 50, 100]

Number of SVD components we train on

Linear classifier

P values: [2, 10, 25, 50, 100]

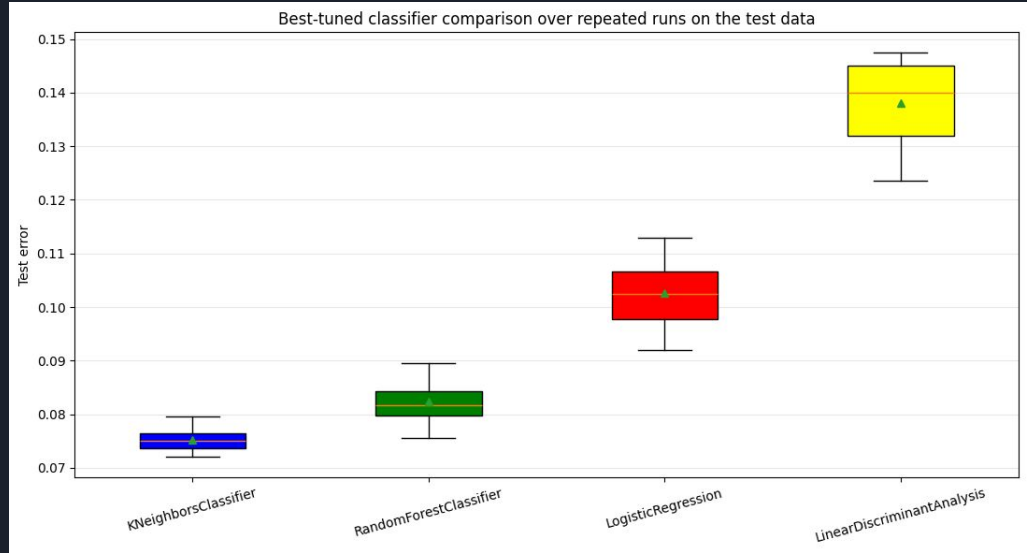
Number of SVD components we train on

Results:

Lowest validation error: KNN with $P=50$ & $k=5$ with 0.07295

Lowest validation error for other models: $P=100$ ($N=200$ for RFC)

Test error for KNN with $P=50$ & $k=5$: 0.0752



Is the data well separated?



Some digits are well separated for the first four features, but in general the data is still clustered. Digits 0 and 1 seems to be the most well separated.

Models required 50-100 features for minimum validation error, indicating that around this many dimensions are required to accurately separate the data.



Part 1 - General discussion

- Significant difference in test error across different models, with the best model (with its best variables) having about half the test error as the worst model (with its best variables)
 - Important to choose the correct model for accurate identification of image data
- Validation error and test error were quite similar, indicating that the validation data represented the test data well
- Std of results across different runs was higher for the worse models, indicating that test error for these depended how the data is split in test, validation and train

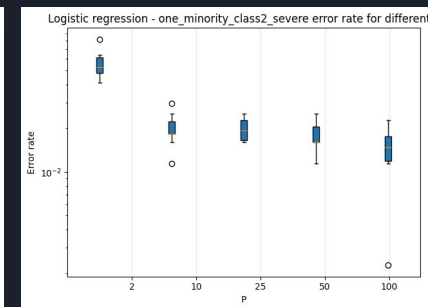
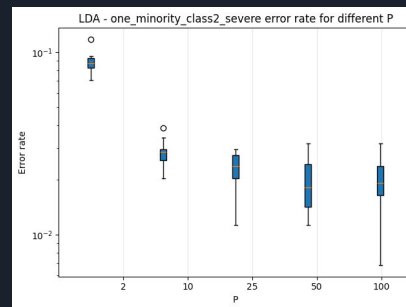
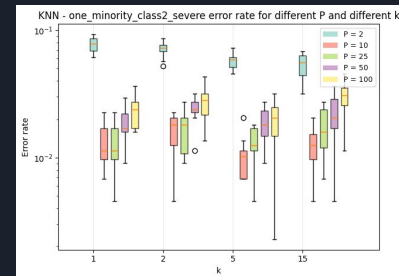
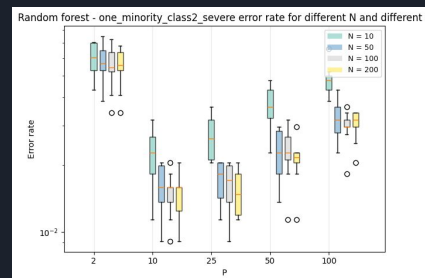
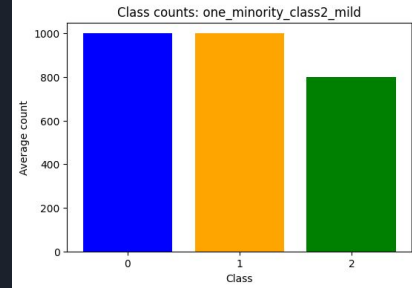
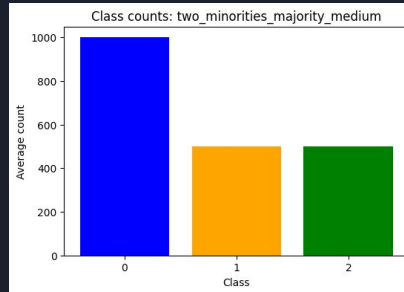


Part 2 - Theme 1: Class imbalance

- We restricted the problem to digits: 0, 1, 2. Then created imbalances in two ways with 4 levels of downsampling; Mild, Medium, Severe, Very severe. We studied class imbalance to see how the classifier behaves when some classes become much rarer than others.
- Case A: Kept two classes unchanged and downsampled the third class.
- Case B: Kept one class unchanged and downsampled the other two classes.
- We evaluated performance using overall accuracy, balanced accuracy, and class-specific accuracy. This was done to understand whether performance is really good.
- The results will show whether the classifier is learning all classes well, or mostly the dominant ones and also shows whether overall accuracy can be trusted in imbalanced settings.
- We wanted to study how the classifiers would behave when the training data is not evenly distributed across classes because in real datasets, some classes often appear much less often than others.

Imbalance scenarios

- With the different scenarios, imbalances becomes stronger from mild to very severe.
- In the one-minority setup, class 2 becomes increasingly rare. In two-minorities setup, class 0 remains the dominant class while 1 and 2 are reduced together.
- This means that classifiers receives much less information about the different classes during training.
- The one minority imbalance mainly tested what happens when one class becomes hard to learn and two minority shows to be harder because now two classes are underrepresented. This makes the classifiers rely even more on the majority class.

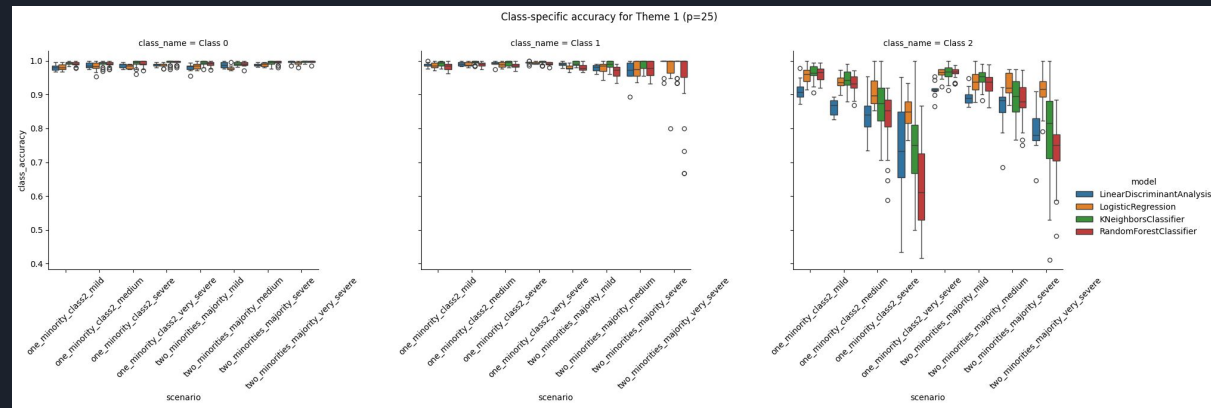




When the data becomes imbalanced

- As imbalance increased, the classifier became worse at recognizing the minority class.
- Overall accuracy often stayed relatively high, especially when the majority class dominated the dataset.
- Balanced accuracy dropped more clearly, because it gives equal weight to each class instead of favoring the largest class.
- Class specific accuracy showed directly which class was suffering most under severe imbalance. This means that a good classifier can still perform poorly where the data is underrepresented.
- The model sees fewer minority class examples during training, so it learns that specific class less accurately.
- Majority classes influence the decision rule more strongly, making minority samples easier to misclassify.
- Overall accuracy alone is not enough when classes are imbalanced.
- Balanced and class specific metrics are necessary to understand real model quality.

Accuracy



- We used different accuracy measures to evaluate the classifier from multiple perspectives and make sure performance was judged fairly, especially under class imbalance.
- Overall accuracy stats high in almost all scenarios but this can be misleading because the majority class dominates the score so the model can look strong even if it performed bad.
- Balanced accuracy drops more clearly as imbalances becomes stronger because it gives equal importances to each class, making it a fair comparison.
- Class specific accuracy shows the real effect of the imbalances most clearly, where class 0 stats almost perfectly classified, Class 1 remains strong but becomes less stable when it is reduced and Class 2 is affected the most. This happens because the classifiers learns best from the classes it sees most often, so when CClass 2 is heavily downsampled, the model has less information about that class and becomes more biased toward the larger classes.
- This shows that class imbalances mainly harms the minority classes while leaving the majority classes almost unaffected.



Part 2 - Discussion

- Part 2 shows that class imbalances mainly harms the minority classes while the majority class remains simple to detect and predict. This explains why the overall accuracy can stay high even when the classifier is not performing for all classes.
- We used overall, balanced and class specific accuracy to measure the performance in different ways. Overall accuracy can be misleading in imbalance data because the majority class dominates the score which can make the model show good results even if it performed badly. Balanced accuracy corrects this by giving equal weight to each class and class-specific accuracy shows exactly which class is being learned well for which model.
- The results shows that classifiers do not react equally to the different models. KNN and Logistic regression often handled the imbalances better while random forest and LDA shows more deterioration in the most difficult setting. This makes sense because minority classes provides fewer examples so the classifiers cannot learn their structure as well and Majority classes dominate the learned decision boundary.